Distributional Reinforcement Learning via Moment Matching (MMDQN)

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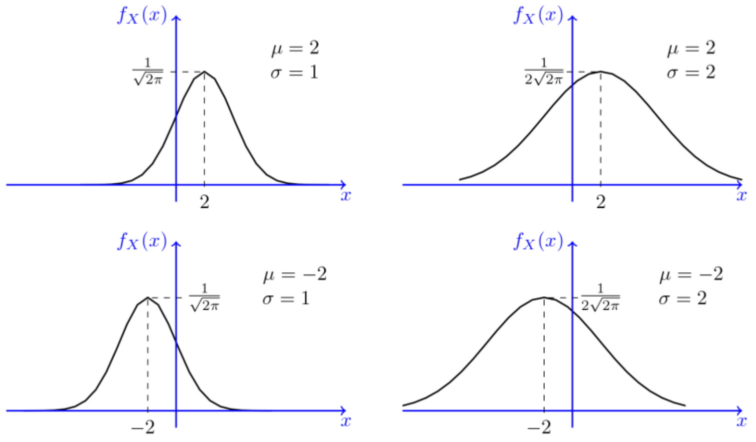
Contents

Introduction

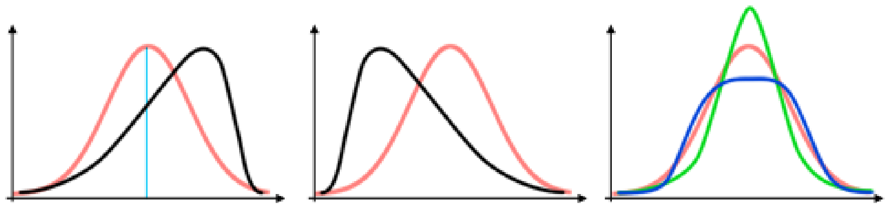
Estimation of the probability distribution

- **Moments of a function are quantitative measures related to the shape of the function's graph**
  - Probability distribution functions(PDF) are generally represented with four measures, mean, variance, skewness, kurtosis
    - Mean : first moment of PDF, ($\mu = E[X]$)
    - Variance : second moment of PDF,($\sigma = E[(X-\mu)^2]^{\frac{1}{2}}$)
    - Skewness : third moment of PDF, ($\gamma = E[\left(\frac{(X-\mu)}{\sigma}\right)^3]$)
    - Kurtosis : fourth moment of PDF, ($Kurt[x] = \frac{E[(X-\mu)^4]}{(E[(X-\mu)^2])^2}$)



Various shape of normal dist. based on μ, σ

- **Estimation methods of PDF $f(x)$ in machine learning**
  - Explicit methods
    - Determination predetermined statistics as any functional $\zeta: f(x) \Rightarrow \mathbb{R}$
      - Median of $f(x)$: $F^{-1}(\frac{1}{2})$, Mean of $f(x)$: $\int_X P(x)dx$
    - Estimation of the predetermined statistics using stacked data
  - Implicit methods
    - Estimation of the $f(x;\theta)$ directly using stacked data(GAN, VAE, …)

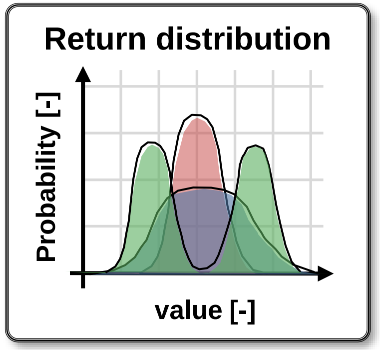


Various shape of normal dist. based on $\gamma, Kurt[x]$

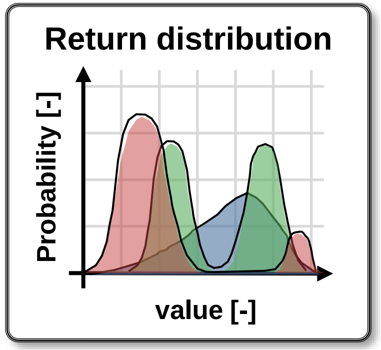
Limitation of distribution estimation in conventional Reinforcement Learning

- **Naïve approaches in estimation of return**
  - Due to stochastic nature of return $G = \Sigma\gamma^t r_{t+1}$, in general RL, G is approximated with $Q(s,a)$ and $V(s)$
    - $V(s) = E[\Sigma\gamma^t r_{t+1} | \, s]$
    - $Q(s,a) = E[\Sigma\gamma^t r_{t+1} | s, a]$
  - These functions, $Q(s,a)$ and $V(s)$ are generally estimated with normal bellman operator $\mathcal{T}_B$ using only mean value
    - $(\mathcal{T}_B \, V)(s) = E[r(s,\pi(s))] + \gamma E[V(s')]$
    - $(\mathcal{T}_B \, Q)(s,a) = E[r(s,a)] + \gamma E[Q(s',a')]$

Variance, skewness, kurtosis of return G are easily neglected



These distributions have same mean, but they are not same!

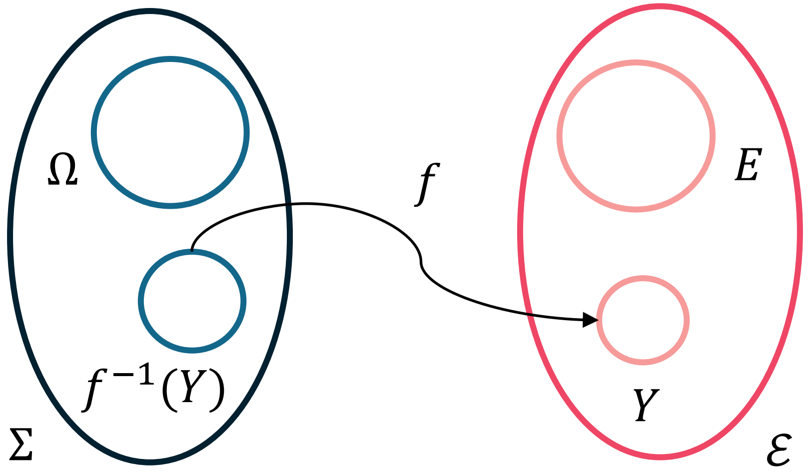


Complex dist. could be modeled in this framework!

Distributional RL via Moment Matching (MMDQN)

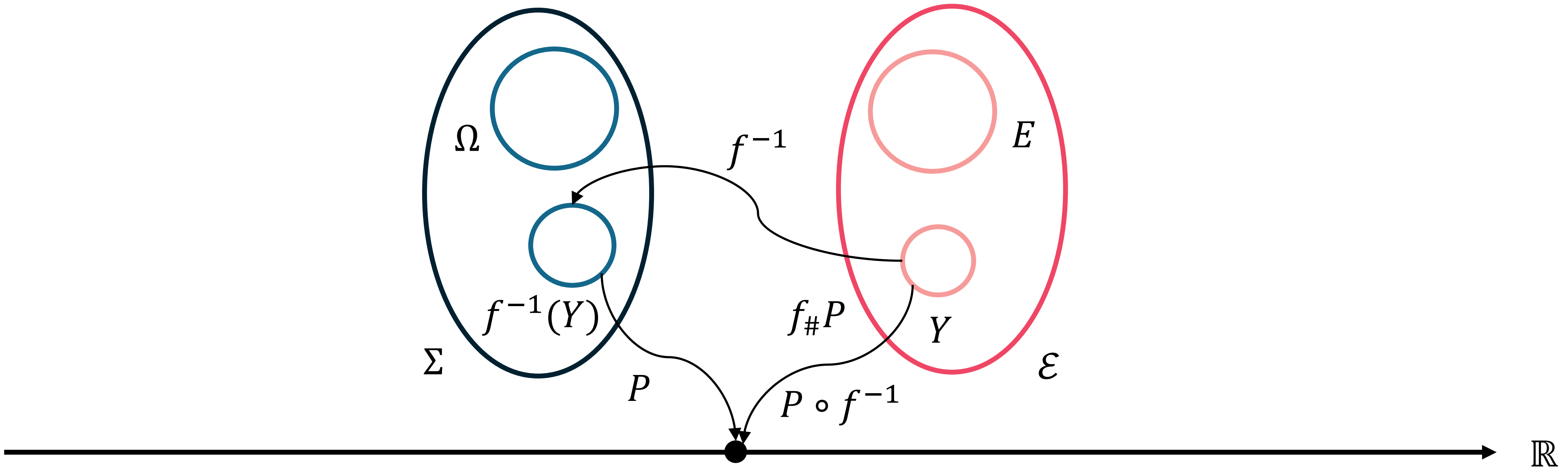
https://arxiv.org/abs/2007.12354

- **Basics of probability space $(\boldsymbol{\Omega}, \boldsymbol{\Sigma}, \boldsymbol{P})$**
  - Sample space Ω
    - In a $probability\ space$, the set Ω is the set of all possible outcomes. Ω is set with element(event) ω, and is called the $sample\ space$
  - $\sigma - algebra\ \Sigma$
    - A $\sigma - algebra\ \Sigma$ is a set of subsets ω of Ω s.t.:
      - $\phi \in \Sigma$
      - If $\omega \in \Sigma, then\ \omega^C \in \Sigma$
      - If $\omega_1, \omega_2, \ldots, \omega_n \in \Sigma, then, U_{i=1}^{\infty}\omega_i \in \Sigma$
      - If $A = \{\phi, \Omega\} \Rightarrow A\ is\ \sigma - algebra,\ A = \{\phi, E, E^C, \Omega\}$ and $E \in \Omega \Rightarrow$ A is $\sigma - algebra$
  - Random variable
    - $B(\mathbb{R})$ is the smallest $\sigma - algebra$ containing open interval I. This set is called Borel set
      - $B(\mathbb{R}) = I = \{(a,b) | a, b \in \mathbb{R}, a < b\}$
    - If f is measurable from (Ω, Σ) to $(E, \mathcal{E})$, it is called a $Borel\ function$ or a $random\ variable$(RV)
      - Let (Ω, Σ) and $(E, \mathcal{E})$ be measurable spaces and f a function from (Ω, Σ) to $(E, \mathcal{E})$. The function f is called $measurable\ function$ iff $f^{-1}(\mathcal{E}) \subset \Sigma$



Backgrounds (II)

- **Basics of probability space $(\boldsymbol{\Omega}, \boldsymbol{\Sigma}, \boldsymbol{P})$**
  - Probability distribution
    - Let (Ω, Σ, μ) be a measure space and f be a measurable function from (Ω, Σ) to $(E, \mathcal{E})$. The pushforward measure is denoted by $f_{\#}\mu$,, $f\#\,\mu$, or $\mu \circ f^{-1}$ is a measure on $\mathcal{E}$ defined as
      - $f_{\#}\mu(Y) = f\#\,\mu(Y) = \mu \circ f^{-1}(Y) = \mu(f \in Y) = \mu\left(f^{-1}(Y)\right), \ \ Y \in \mathcal{E}$
    - If $\mu = P$ is a probability measure, and f is random variable, then $P \circ f^{-1}$ is called the distribution (or the law) of f and is denoted by P_X
      - P is $\mathit{probability\ distribution}$ or $\mathit{probability\ measure}$
      - In practice, one often use $(E, \mathcal{E}) = \left(\mathbb{R}^n, B(\mathbb{R}^n)\right)$, $f\#\mu: B(\mathbb{R}) \to [0, 1]$



Backgrounds (III)

- **Distributional RL**
  - In distributional RL, the cumulative return of a chosen action at state is modeled with the full distribution rather than expectation of it, $Z_\theta(s,a) \coloneqq \frac{1}{N}\Sigma_{i=1}^{N}\delta_{\theta_i}(s,a)$
    - So that the model can capture its intrinsic randomness instead of just first-order moment(high-order moments, multi-modality in state-action value function)
  - Existing distributional RL algorithms
    - **C51:** Pre-determination of the supports of return distribution and then training the categorical distribution.
    - **QR-DQN:** To avoid pre-determined supports, and decrease the theory-practice gap, introducing the quantile regression
    - **IQN:** Using sampling the quantile, training the full quantile function, and considering the risk of policy
    - **IDAC**: Training the full return distribution directly using the adversarial network, and training the policy based on semi-implicit methods

MMDQN (I)

- **Overview of the MMDQN**
  - Unlike the existing distributional RL, MMDQN has no assumption about predetermined statistics and could learn the unrestricted statistics
    - However, for implementation, deterministic pseudo-samples of the return distribution are learned in MMD
      - The authors use the Dirac mixture $\hat{\mu}_\theta(s,a) = \frac{1}{N}\Sigma_{i=1}^{N}\delta_{Z_\theta(s,a)}$ to approximate $\mu^\pi(s,a)$

  - The authors analyze the distributional Bellman operator and establish sufficient conditions for the contraction of the distributional Bellman operator in the MMD
    - They analyzed the $\mathcal{T}^\pi$ is a contraction when the kernel is unrectified kernel $k_\alpha(x,y) \coloneqq -\left||x-y|\right|^\alpha, \forall\alpha \in \mathbb{R}, \forall x,y \in \mathcal{X}$
    - However, for practical consideration, they demonstrates the commonly used Gaussian kernel has better performance in this framework($k(x,y) \coloneqq \exp\left(-\frac{(x-y)^2}{2\sigma^2}\right)$)

  - MMDQN with Gaussian kernel mixture showed the state-of-the-art in the 55 Atari 2600 games.
    - For a fair comparison, they used the same architecture of DQN and QR-DQN

MMDQN (II)

- **Problem setting**
  - For any policy π, let $\mu^{\pi} = \text{law}(Z^{\pi})$ be the distribution of the return RV. $Z^{\pi}(s,a) \coloneqq \Sigma_{t=0}^{\infty} \gamma^{t} R(s_t, a_t)$
  - $\mathcal{T}^{\pi}\mu(s,a) \coloneqq \int_{\mathcal{S}}\int_{\mathcal{A}}\int_{\chi} (f_{\gamma,r})\#\mu(s',a')\mathcal{R}(dr|s,a)\pi(da'|s)P(ds'|s,a)\,, \quad f_{\gamma,r}(z) \coloneqq r + \gamma z, \quad \forall z \; and$

  $(f_{\gamma,r})\#\mu(s',a')$ is the push forward measure of $\mu(s',a')$ *by* $f_{\gamma,r}$

- **Algorithmic approach**
  - The authors use the Dirac mixture $\hat{\mu}_{\theta}(s,a) = \frac{1}{N}\Sigma_{i=1}^{N}\delta_{Z_{\theta}(s,a)}$ to approximate $\mu^{\pi}(s,a)$
    - They referred to the deterministic samples $Z_{\theta}(s,a)$ as particles
  - Algorithm goal is reduced into learning the particles $Z_{\theta}(s,a)$ to approximate $\mu^{\pi}(s,a)$.
    - To this end, the particles $Z_{\theta}(s,a)$ is deterministically evolved to minimize the MMD distance between the approximate distribution and its distributional Bellman target

MMDQN (II)

- **Maximum Mean Discrepancy(MMD)**
  - Let $\mathcal{F}$ be a Reproducing Kernel Hilbert Space(RKHS) associated with a continuous kernel $k(\cdot,\cdot)$ on $\mathcal{X}$.
  - The MMD between $p \in P(\mathcal{X})$ and $q \in P(\mathcal{X})$ is defined as
    - $\mathrm{MMD}(p, q; \mathcal{F}) \coloneqq \sup_{f\in\mathcal{F}:||f||\le 1} (\mathbb{E}[\mathrm{f}(\mathrm{Z})] - \mathbb{E}[f(W)]) = \left|\left|\int_{\mathcal{X}} k(x,\cdot)p(dx) - \int_{\mathcal{X}} k(x,\cdot)q(dx)\right|\right|_{\mathcal{F}}$

      $= (\mathbb{E}[k(Z,Z')] + \mathbb{E}[k(W,W')] - 2\mathbb{E}[k(Z,W])^{\frac{1}{2}},\ \ Z, Z' \sim p, W, W' \sim q$ and they are independent respectively.

  - In practical, MMD is biased estimated from $\mathrm{MMD_b}$ with empirical samples $\{z_i\}_{i=1}^N \sim p$ and $\{w_i\}_{i=1}^M \sim q$.
    - $\mathrm{MMD}_\mathrm{b}^2(\{z_i\},\{w_i\}; k) = \frac{1}{N^2}\Sigma_{i,j} k\left(z_i, z_j\right) + \frac{1}{M^2}\Sigma_{i,j} k\left(w_i, w_j\right) - \frac{2}{NM}\Sigma_{i,j} k(z_i, w_j)$

  - The authors utilized the Gaussian kernel $k(x, y) = \exp\left(-\frac{|x-y|^2}{h}\right)$ for objective $\mathrm{MMD_b}$
    - They exemplified the following intuition:
      - The first term serves as a repulsive force that pushes the particles $\{Z_\theta(s,a)_i\}$ away from each other, preventing them from collapsing into a single mode
      - The third term acts as an attractive force which pulls the particle $\{Z_\theta(s,a)_\mathrm{i}\}$ closer to their target particles $\{\hat{\mathcal{T}} Z_i\}$.

    - They used the kernel mixture trick with K kernels(they are different in bandwidth h)

MMDQN (III)

- **Pseudo code**
  - Hyper-params inputting and target network initialization corresponds with the main network

  - Every single step, transition is sampled from the replay buffer
    - In the control setting, action is inferred from the policy($\epsilon - greedy$)
    - In the policy evaluation, action is selected with the estimated return distribution $\hat{\mu}(s', a')$

  - For the number of statistics N, target return value $\hat{T}Z_i$ is computed

  - $\mathrm{MMD_b}$ objective is computed and backpropagated with SGD

Algorithm 1: Generic MMDRL update

Require: Number of particles N, kernel k, discount factor $\gamma \in [0, 1]$

Input: Sample transition (s, a, r, s')

if *Policy evaluation* **then**

  $a^* \sim \pi(\cdot|s')$

end

else if *Control setting* **then**

  $a^* \leftarrow \arg\max_{a' \in \mathcal{A}} \frac{1}{N} \sum_{i=1}^{N} Z_\theta(s', a')_i$

end

$\hat{T}Z_i \leftarrow r + \gamma Z_{\theta^-}(s', a^*)_i, \forall 1 \leq i \leq N$

Output: $\mathrm{MMD}_b^2 \left(\{Z_\theta(s, a)_i\}_{i=1}^{N}, \{\hat{T}Z_i\}_{i=1}^{N}; k \right)$

Pseudo code of MMDQN

Experiment Results

Experiment results (I)

- **Comparison with previous methods**

*HN scores: Human Normalized score

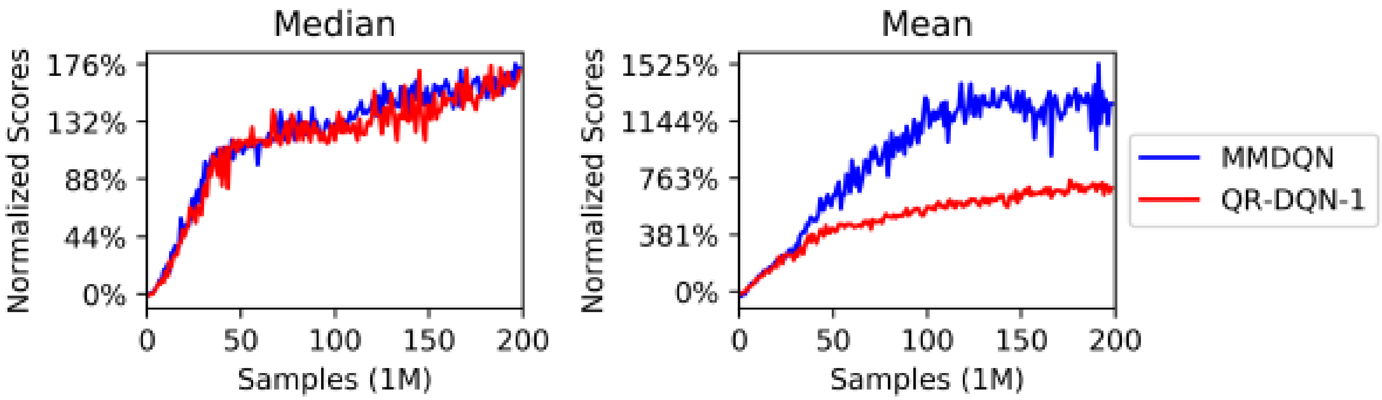
  - Experiments show the superior performance of MMDQN compared to previous methods in 55 Atrai 2600 games(OpenAI Gym env)

- DQN
- C51
- PRIOR
- QR-DQN

➔ First group

- RAINBOW
- FQF
- IQN

➔ Second group



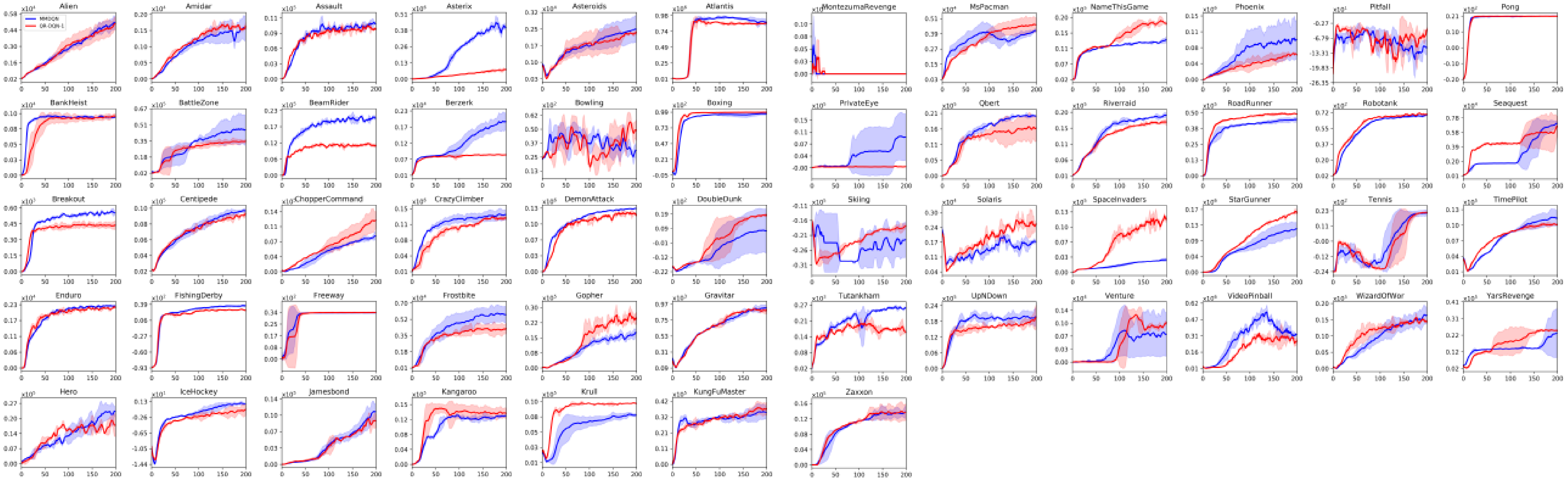
	Mean	Median	>Human	>DQN
DQN	221%	79%	24	0
PRIOR.	580%	124%	39	48
C51	701%	178%	40	50
QR-DQN-1	902%	193%	41	54
RAINBOW	1213%	227%	42	52
IQN	1112%	218%	39	54
FQF	1426%	272%	44	54
MMDQN	**1969%**	213%	41	**55**

Median and mean of the *HN scores

Median and mean of best *HN scores

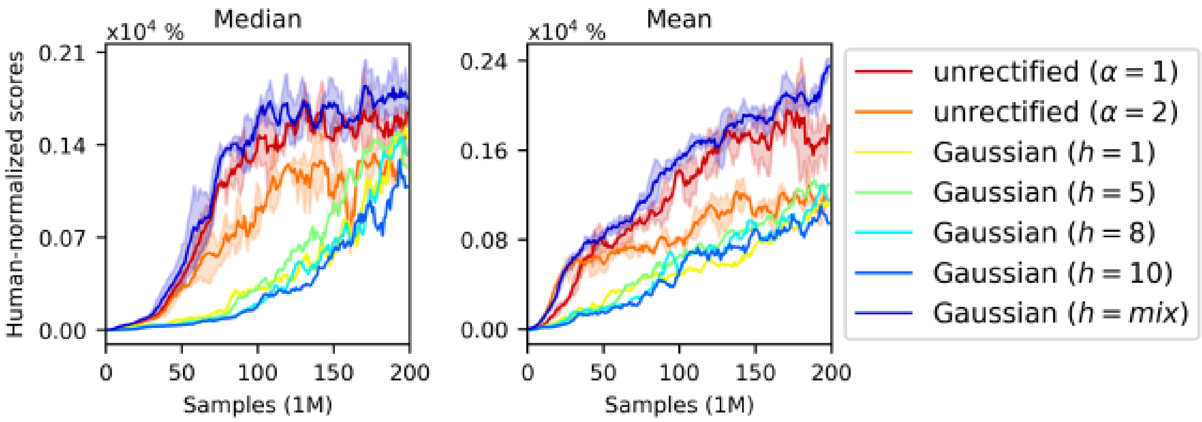
Experiment results (III)

- **Comparison with the previous method**
  - Experiments show the superior performance of MMDQN compared to the previous method(QR-DQN) in 55 games in Atari 2600 games(OpenAI Gym env)
    - MMD(3 seeds)
    - QR-DQN(2 seeds)

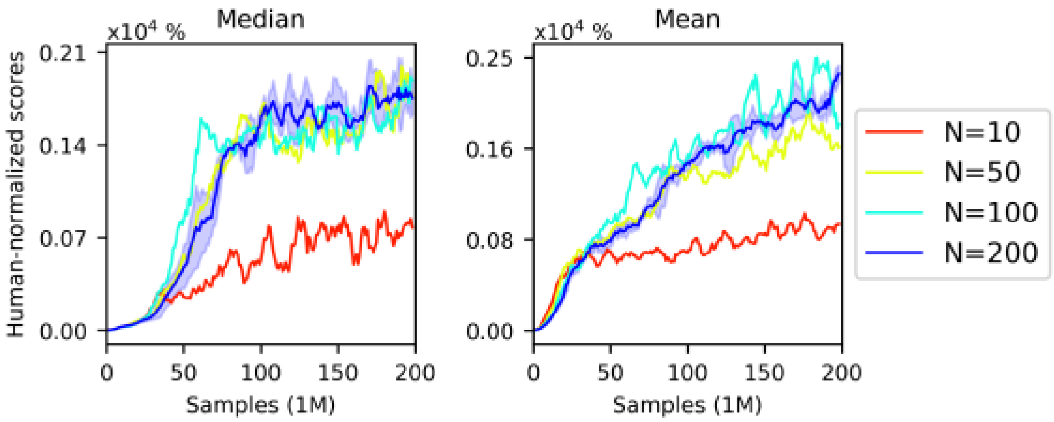


Online training curves for MMDQN and QR-DQN

- **Ablation study**
  - Two sets of ablation studies were performed to answer the following questions
    - (a)**:** Which kernel used for the MMDQN shows the best performance?
      - Using the mixture of Gaussian kernels with different bandwidths displays the best performance
    - (b)**:** What number of particles for the MMDQN shows the best performance?
      - Using more than 50 particles of dist. demonstrates better performance, and using 200 particles of dist. exibits the most s table performance



(a) Different kernel bandwidths (at fixed $N = 200$).



(b) Different values of N (at fixed $h = mix$).

The sensitivity of MMDQN in the 6 tuning games w.r.t (a): the kernel choice, and (b): the number of particles N

Thank you!

Q&A

Appendix

- **Basics of measure theory**
  - Measurable space (X, Σ)
    - A pair (X, Σ) is a $measurable\ space$ if X is a set and Σ is a nonempty $\sigma - algebra$ of subsets of X
    - A measurable space allows us to define a function that assigns real numbered values to the abstract elements of Σ
  - Measure μ
    - Let (X, Σ) be a measurable space, set function μ defined on Σ is called $measure$ iff has the following properties
      - $0 \leq \mu(A) \leq \infty\ for\ any\ A \in\ \Sigma$
      - $\mu(\Phi) = 0$
      - For any sequence of pairwise disjoint sets $\{A_n\} \in \Sigma$ such that $U_{n=1} A_n \in \Sigma$, we have $\mu(U_{n=1}^{\infty} A_n) = \Sigma_{n=1}^{\infty} \mu(A_n)$
  - Measure space
    - A triplet (X, Σ, μ) is a $measure\ space$ if (X, Σ) is a $measurable\ space$ and $\mu: \Sigma \rightarrow [0; \infty)$ is a $measure$
    - If $\mu(X) = 1$, then μ is a $probability\ measure$, which we usually use notation P, and the measure space is a $\boldsymbol{probability\ space}$

- **Basics of measure theory**
  - Measurable function
    - Let (Ω, Σ) and (Λ, G) be measurable spaces and f a function from Ω to Λ. The function f is called $measurable\ function$ from (Ω, Σ) to (Λ, G) iff $f^{-1}(G) \subset \Sigma$
  - Random variable
    - A random variable $\boldsymbol{X}$ is a measurable function from the probability space (Ω, Σ, P) into the probability space $(\mathcal{X}, B_{\mathcal{X}}, P_{\mathcal{X}})$, where $\mathcal{X}$ in $\mathbb{R}$ is the range of the $\boldsymbol{X}$, $B_{\mathcal{X}}$ is a $Borel\ set\ of\ \mathcal{X}$ and $P_{\mathcal{X}}$ is the probability measure(distribution) on $\mathcal{X}$
      - Specifically, $\boldsymbol{X}: \Omega \rightarrow \mathcal{X}$